

GroupTrack

User Manual

Version 2.4

Off-Road Convoy Tracking for Android

May 2026

1. What is GroupTrack?

GroupTrack is an Android application for off-road convoy tracking. It uses Meshtastic mesh radios to share real-time GPS positions between riders in a group, allowing everyone to see where the convoy is on an offline map — even in areas with no cell service.

GroupTrack shows each rider's position on the map in real time, identifies the lead and tail vehicles, records GPS tracks of your ride, and works completely offline using pre-downloaded map tiles.

1.1 What You Need

Item	Details
Android device	Android 10 or later. Tested on Samsung and Pixel devices.
Meshtastic radio	T1000-E or compatible Meshtastic device. Must be on the same frequency and channel as other riders.
Bluetooth	The app connects to the radio via Bluetooth Low Energy (BLE).
Map tiles (optional)	Download offline map tiles over WiFi before your ride. Without tiles, the map shows a blank background.

⚠ Budget Android tablets (Tab8NEU and similar) have known Bluetooth issues with T1000-E radios. The BLE connection drops after 8–10 seconds. Use a name-brand phone or tablet for reliable operation.

2. Getting Started

2.1 Install GroupTrack

If you received a Google Play internal test invite, follow the installation instructions provided by your ride organizer. If you received a direct APK file, see the side-loading instructions in the install guide.

2.2 Grant Permissions

GroupTrack requires several permissions to function. The app will prompt you for each one on first launch.

Location Permission

1. When prompted, tap "Allow While Using the App" first.
2. The app will then explain why it needs background location access and ask you to grant "Allow All the Time."
3. This opens your device Settings. Find GroupTrack in the list and select "Allow all the time."
4. Return to the app.

⚠ If you only grant "While using the app," GPS track recording will stop when you turn off your screen or switch apps. Your ride track will be empty.

Storage Permission

1. The app will request "All Files Access" permission.
2. This opens your device Settings. Find GroupTrack and toggle on "Allow access to manage all files."
3. Return to the app.

This permission is required for offline map tiles and track file storage.

2.3 Connect to Your Radio

1. Power on your Meshtastic radio (T1000-E or compatible).
2. In GroupTrack, you will see a Bluetooth connection prompt or a radio list.
3. Select your radio from the list. The app connects via BLE.
4. Once connected, your position and other riders' positions will appear on the map.

✓ Keep your radio powered on and within Bluetooth range of your phone (typically 10–30 feet) throughout the ride.

3. The Convoy Map

The main screen is the convoy map. It shows your position and every connected rider in the group.

3.1 Map Display

- Your cart ("My Cart") is shown on the map with your position updated in real time.
- Other riders in the convoy appear as separate markers. Each rider's position updates as the mesh radio network relays GPS data.
- The map uses hybrid satellite/terrain tiles by default.
- Trail overlays (GeoJSON) display known OHV trails when available.

3.2 Lead and Tail

GroupTrack automatically identifies the lead vehicle (front of convoy) and the tail vehicle (rear of convoy). The lead cart is set when you start a group track. The convoy engine computes positions and heading relative to the lead.

3.3 Speed Display

Your current speed (mph) is displayed on the map HUD after a brief warmup period. During the first 60 seconds, the speed display may show 0 mph while the GPS acquires enough position samples to compute accurate speed.

⚠ Mileage traveled (trip odometer) is not displayed in V2.4. This feature is planned for a future release.

4. Recording a Ride

4.1 Start Recording

1. Tap the RECORD button on the map screen.
2. If prompted for location permission, grant "Allow all the time" as described in Section 2.2.
3. The app begins recording your GPS track. A recording indicator appears on screen.
4. The lead cart is set automatically (your radio node, or "My Cart" in phone-only mode).

4.2 During the Ride

- Your GPS position is recorded every 3 seconds and saved to a track file.
- The track line draws on the map as you move.
- Other riders' positions update via the mesh radio network.
- You can lock and unlock your phone — recording continues in the background (requires "Allow all the time" location permission).

4.3 Stop Recording

1. Tap the STOP button.
2. A dialog appears asking you to name the track.
3. Enter a name (e.g., "Bar 10 Trail April 2026") and confirm.
4. The track is saved as a GPX file in your device storage.

4.4 Viewing Saved Tracks

Tap the track list button to see all recorded tracks. Select any track to view it as an overlay on the map. Tracks are stored in the Documents/my_tracks/ folder and can be exported via the system file manager.

5. Offline Map Tiles

GroupTrack works best with pre-downloaded map tiles. Without them, the map shows a blank background in areas with no cell service.

5.1 Downloading Tiles

1. Connect to WiFi before your ride.
2. On the map screen, navigate to the area you want to download.
3. Use the tile download controls to select a region.
4. Tap DOWNLOAD. The app downloads satellite/hybrid tiles for the selected area at multiple zoom levels.
5. Wait for the download to complete. Large areas may take several minutes.

✓ Download tiles the night before your ride when you have a strong WiFi connection. A typical ride area (20–30 miles of trail) requires 200–500 MB of tile storage.

5.2 Viewing Downloaded Areas

Tap the SHOW DOWNLOADED button to see which areas have cached tiles. The downloaded region is highlighted on the map. A progress indicator shows while the scan runs — the button is disabled during the scan to prevent duplicate operations.

6. Radio Configuration

GroupTrack includes tools to configure your Meshtastic radio for convoy use. The recommended settings are LONG_FAST mode, which provides 3–5 mile range with good data throughput for position updates.

- **Apply Radio:** Applies the convoy master configuration to your connected radio.
- **Archive/Restore:** Backs up your current radio settings before applying convoy config, and restores them after the ride.

⚠ All riders in a convoy must be on the same frequency, channel, and modem preset. Use the Apply Radio feature to ensure consistent configuration.

7. Known Limitations (V2.4)

Limitation	Details
No mileage display	Trip distance / odometer is not calculated or displayed. Planned for a future release.
60s speed warmup	Speed shows 0 mph for the first ~60 seconds while GPS acquires enough samples.
GPS interval 3s	Track points are recorded every 3 seconds. Higher resolution (1s) is planned for V2.5.
Budget device BLE	Low-cost tablets (Tab8NEU, etc.) cannot hold BLE connection to T1000-E radios. Use a name-brand Android phone.
BLE reconnect	If BLE drops during a ride, manually disconnect and reconnect in the app. Workaround: keep phone close to radio.
Battery optimization	Some devices kill background apps aggressively. Go to Settings → Battery → GroupTrack → Unrestricted to prevent this.

8. Troubleshooting

Track file is empty (0 track points)

Location permission is set to "While using the app" instead of "Allow all the time." Go to Settings → Apps → GroupTrack → Permissions → Location → Allow all the time.

Map is blank / no tiles visible

Offline tiles have not been downloaded for this area. Connect to WiFi and use the tile download feature before your ride. Also confirm that All Files Access permission is granted.

Radio won't connect

Ensure the radio is powered on and within Bluetooth range. Try toggling Bluetooth off and on in your phone's Settings. If using a budget tablet, try a different Android device — see the BLE compatibility note in Section 7.

Speed stuck at 0 mph

Wait 60 seconds after starting a ride. The speed display requires a warmup window to compute accurate speed from GPS samples. If speed remains at 0 after 2 minutes, check that location permission is set to "Allow all the time."

App killed in background

Go to Settings → Battery → find GroupTrack → set to "Unrestricted." This prevents Android from killing the app when the screen is off.

